Lab Assignment # 08

Course Title : AI Assistant Coding Name
of Student : N.Ananthan
Enrollment No. : 2303A54062
Batch No. : 47_B

Lab 8: Test-Driven Development with AI – Generating and Working with Test Cases

Task Description #1 (Password Strength Validator – Apply AI in Security Context)
• Task: Apply AI to generate at least 3 assert test cases for is_strong_password(password) and implement the validator function.

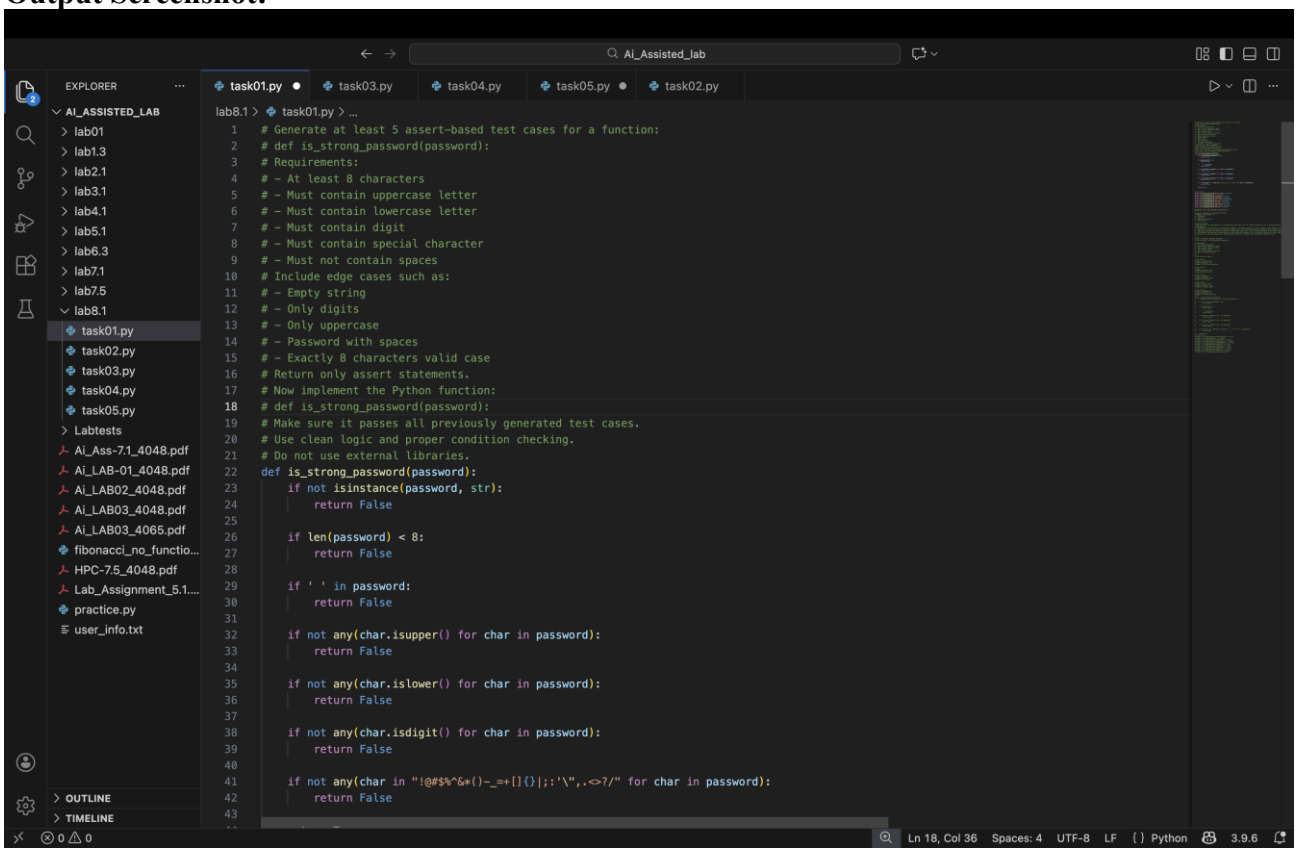
• Requirements:
o Password must have at least 8 characters.
o Must include uppercase, lowercase, digit, and special character.
o Must not contain spaces.

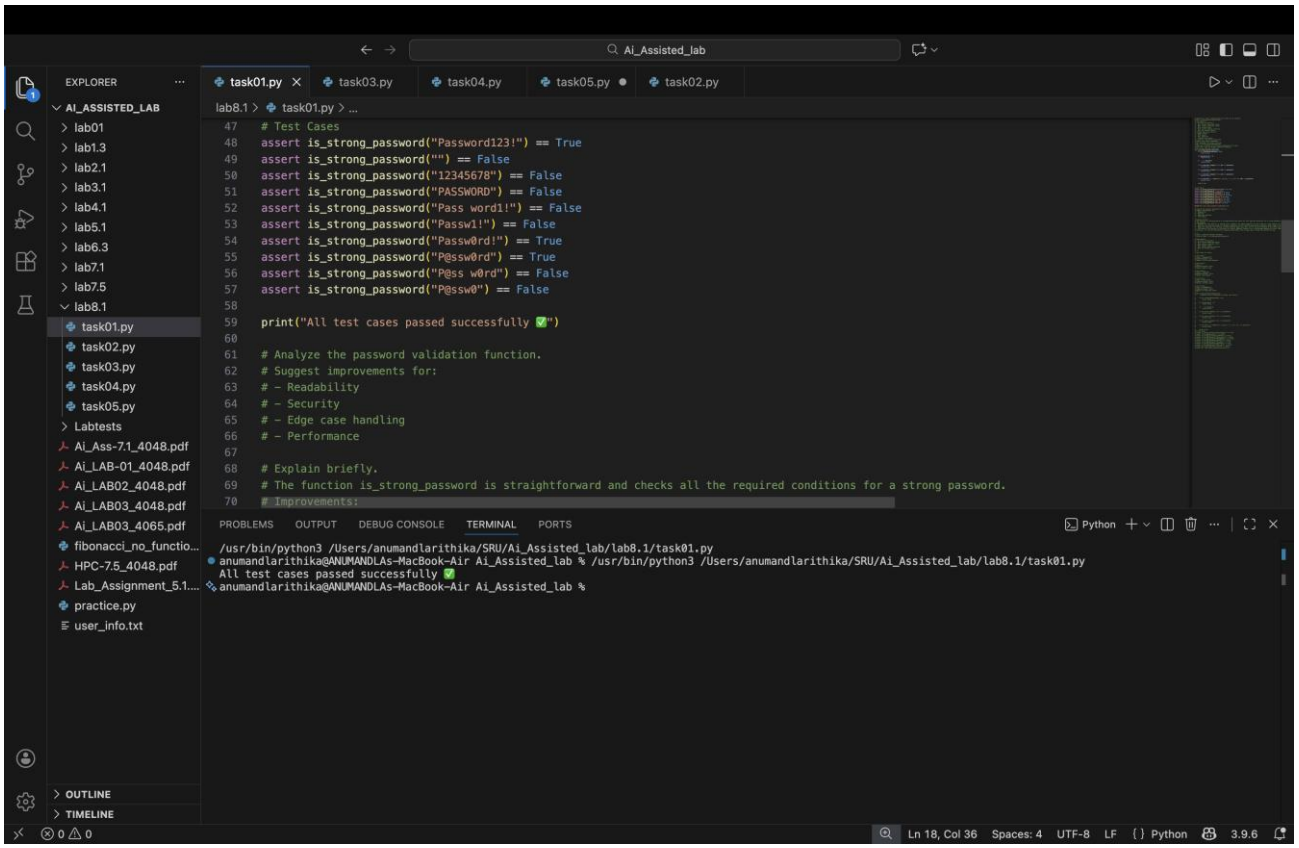
Example Assert Test Cases:

```
assert is_strong_password("Abcd@123") == True
assert is_strong_password("abcd123") == False
assert is_strong_password("ABCD@1234") == True
```

Expected Output #1: • Password validation logic passing all AI-generated test cases.

Output Screenshot:





Explanation: This checks if a password is truly "strong" by making sure it has the right mix of characters and no spaces.

Task Description #2 (Number Classification with Loops – Apply AI for Edge Case Handling)

• Task: Use AI to generate at least 3 assert test cases for a classify_number(n) function. Implement using loops.

• Requirements:

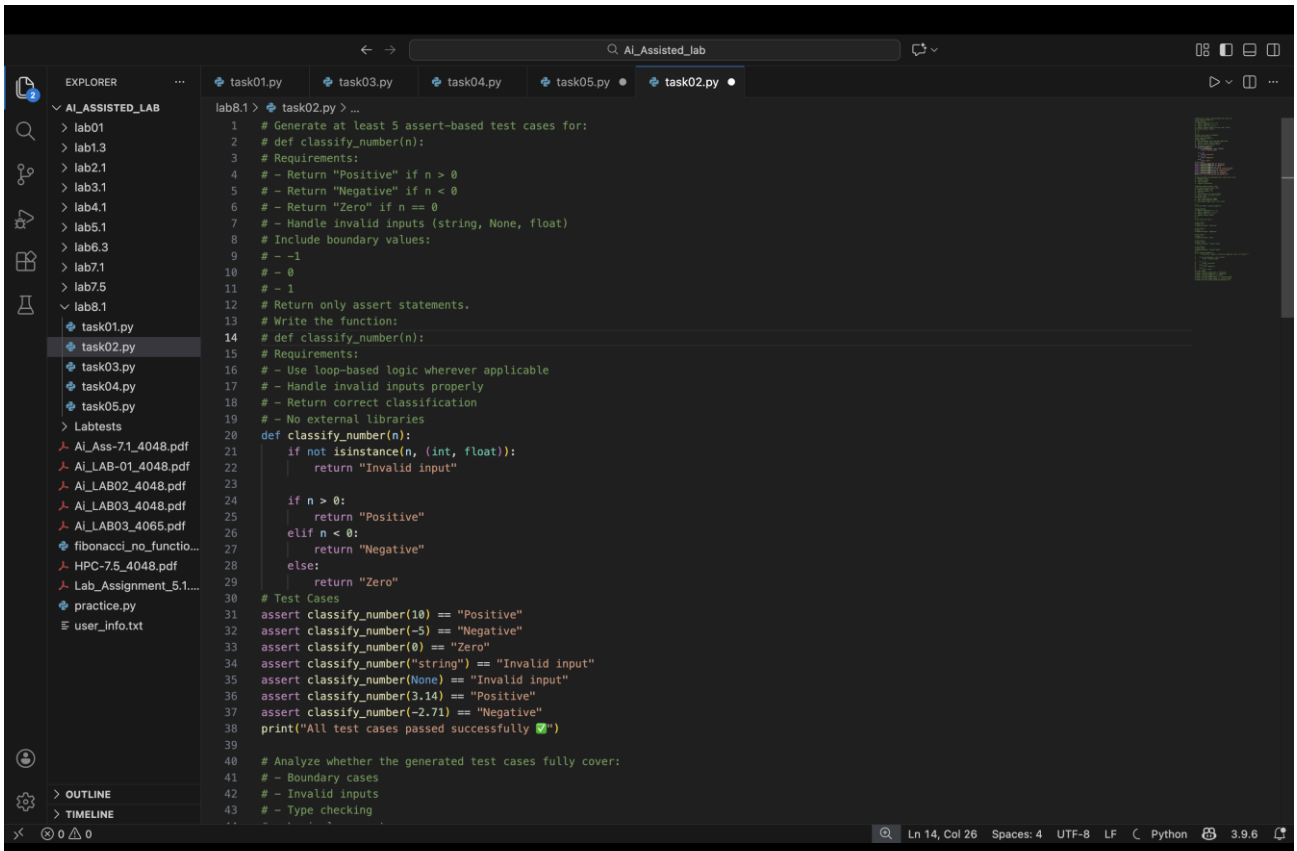
- o Classify numbers as Positive, Negative, or Zero.
- o Handle invalid inputs like strings and None.
- o Include boundary conditions (-1, 0, 1).

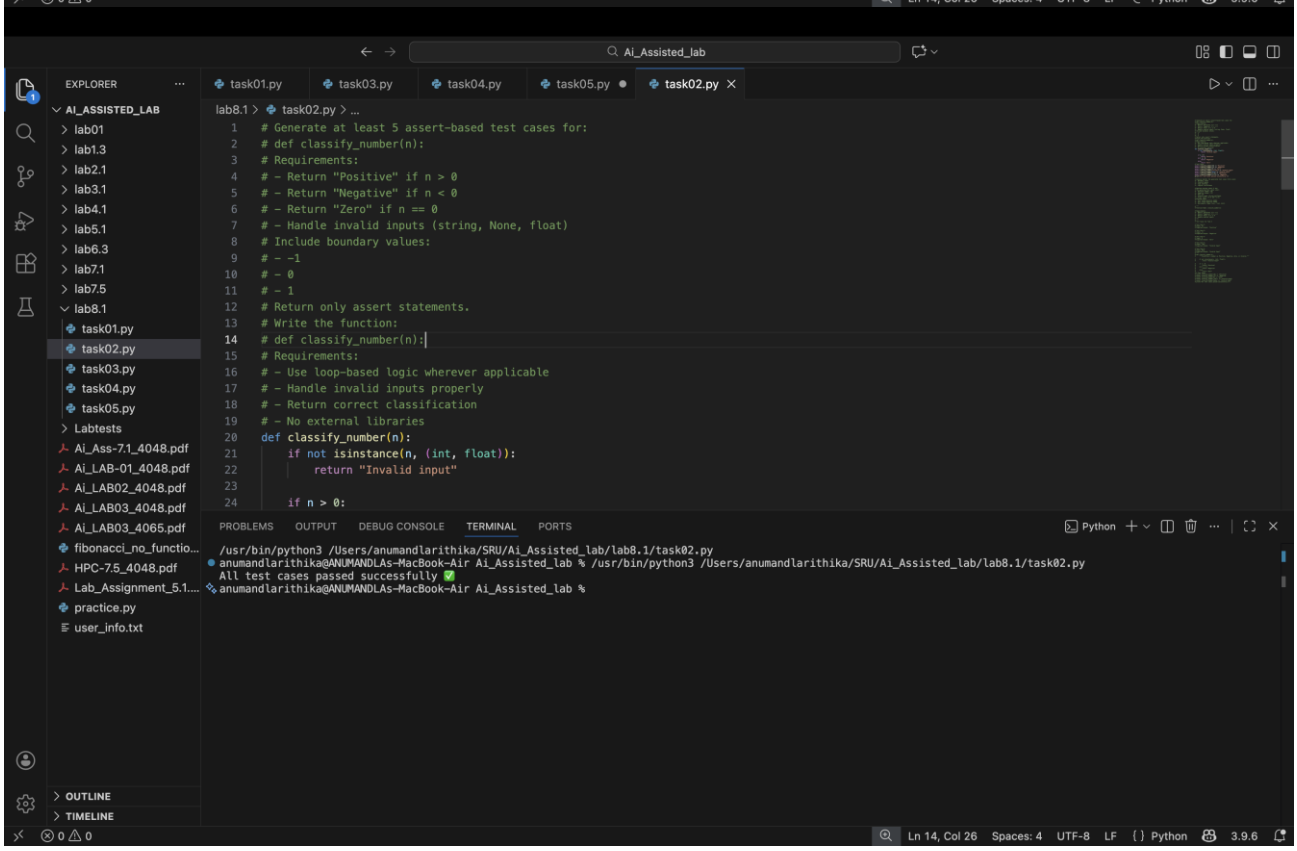
Example Assert Test Cases:

```
assert classify_number(10) == "Positive"
assert classify_number(-5) == "Negative"
assert classify_number(0) == "Zero"
```

Expected Output #2: • Classification logic passing all assert tests.

Output Screenshot:





Explanation: This sorts numbers into positive, negative, or zero, while politely rejecting anything that isn't a number.

Task Description #3 (Anagram Checker – Apply AI for String Analysis)

• Task: Use AI to generate at least 3 assert test cases formis_anagram(str1, str2) and implement the function.

• Requirements:

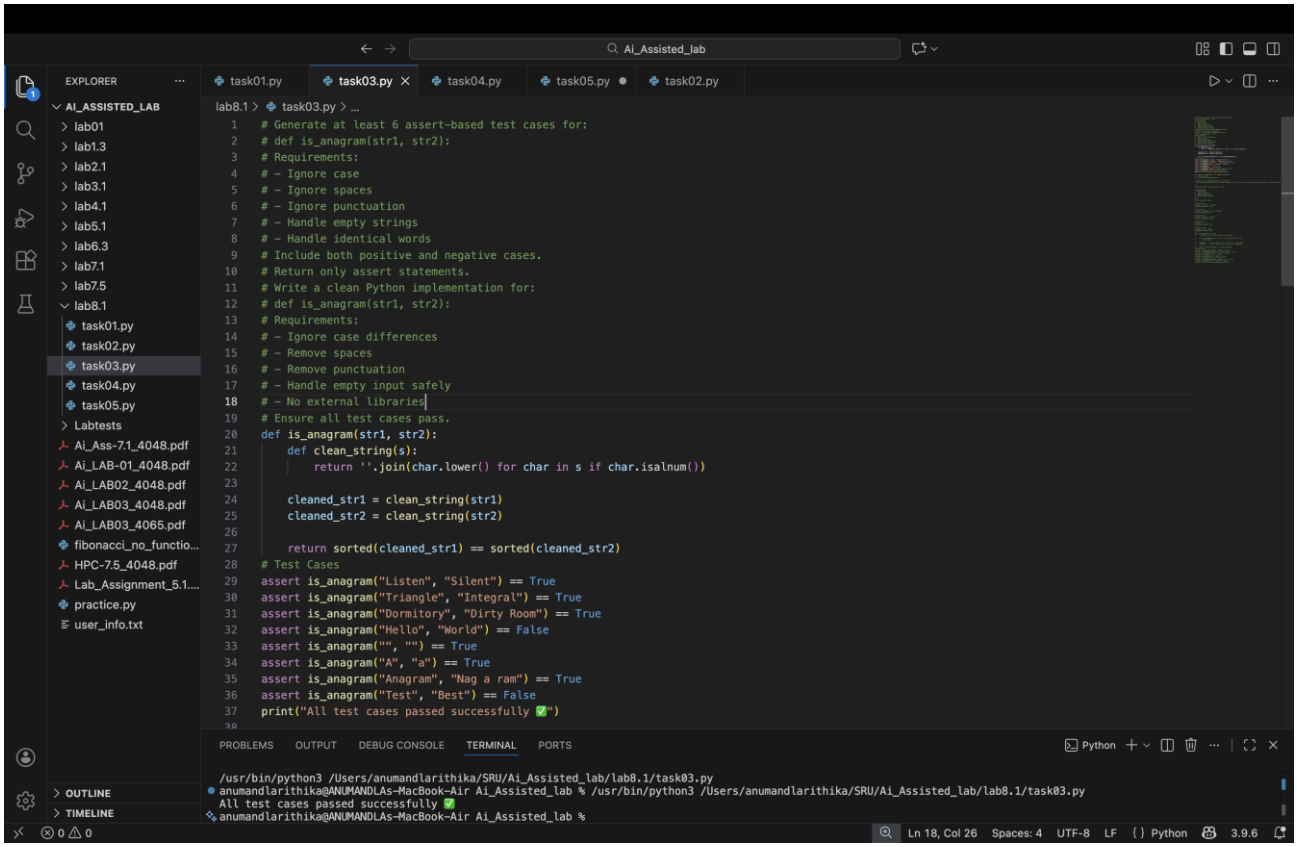
o Ignore case, spaces, and punctuation.
o Handle edge cases (empty strings, identical words).

Example Assert Test Cases:

assert is_anagram("listen", "silent") == True
assert is_anagram("hello", "world") == False
assert is_anagram("Dormitory", "Dirty Room") == True

Expected Output #3: • Function correctly identifying anagrams and passing all AI-generated tests.

Output Screenshot:



Explanation: This spots whether two words or phrases are made of the same letters, ignoring case, spaces, and punctuation.

Task Description #4 (Inventory Class – Apply AI to Simulate Real- World Inventory System)

• Task: Ask AI to generate at least 3 assert-based tests for an Inventory class with stock management.

• Methods:

o add_item(name, quantity)
o remove_item(name, quantity)
o get_stock(name)

Example Assert Test Cases:

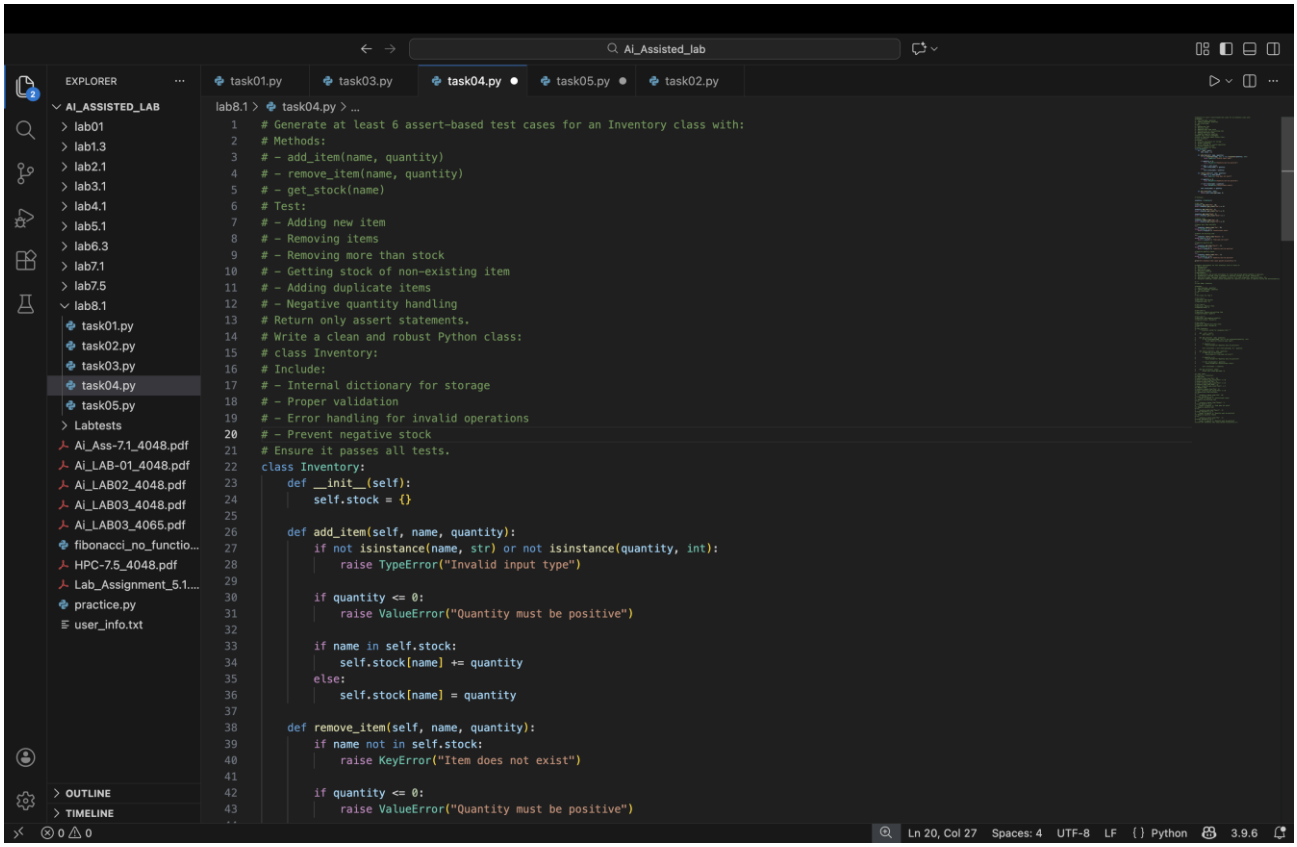
inv = Inventory()

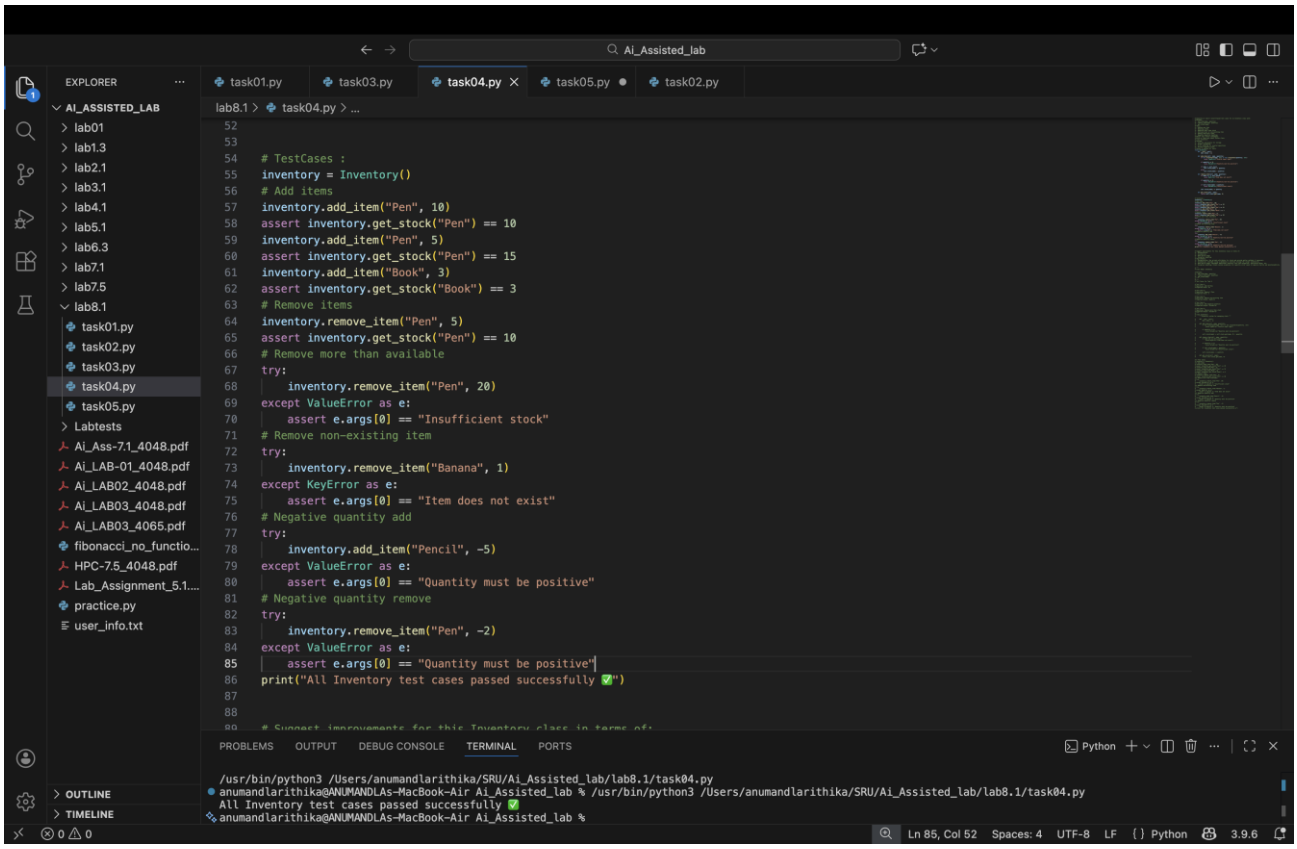
inv.add_item("Pen", 10)
assert inv.get_stock("Pen") == 10

```
inv.remove_item("Pen", 5)
assert inv.get_stock("Pen") == 5
inv.add_item("Book", 3)
assert inv.get_stock("Book") == 3
```

Expected Output #4: • Fully functional class passing all assertions.

Output Screenshot:





Explanation: This simulates a mini store system where you can add, remove, and check stock for items.

Task Description #5 (Date Validation & Formatting – Apply AI for Data Validation)

• Task: Use AI to generate at least 3 assert test cases for validate_and_format_date(date_str) to check and convert dates.

• Requirements:

- o Validate "MM/DD/YYYY" format.
- o Handle invalid dates.
- o Convert valid dates to "YYYY-MM-DD".

Example Assert Test Cases:

assert validate_and_format_date("10/15/2023") == "2023-10-15"
assert validate_and_format_date("02/30/2023") == "Invalid Date"
assert validate_and_format_date("01/01/2024") == "2024-01-01"

Expected Output #5: • Function passes all AI-generated assertions and handles edge cases.

Output Screenshot:

```python
# Generate at least 6 assert test cases for:
# def validate_and_format_date(date_str):
# Requirements:
# - Input format must be MM/DD/YYYY
# - Validate correct calendar date
# - Handle leap year
# - Reject invalid format
# - Return "Invalid Date" for invalid inputs
# - Convert valid date to YYYY-MM-DD
# Include boundary cases:
# - 02/29/2024 (leap year)
# - 02/29/2023 (invalid)
# - 13/01/2023 (invalid month)
# - 00/10/2023
# Return only assert statements.
# Write the function:
# def validate_and_format_date(date_str):
# Requirements:
# - Validate format manually (no external libraries)
# - Check month range
# - Check correct number of days
# - Handle leap years properly
# - Return formatted date if valid
# - Else return "Invalid Date"
# Keep the code clean and readable.
def validate_and_format_date(date_str):
    def is_leap_year(year):
        return (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0)

    try:
        month, day, year = map(int, date_str.split('/'))
    except ValueError:
        return "Invalid Date"

    if month < 1 or month > 12:
        return "Invalid Date"

    if day < 1:
        return "Invalid Date"

    if month in [1, 3, 5, 7, 8, 10, 12]:
        if day > 31:
            return "Invalid Date"
```

```python
def validate_and_format_date(date_str):
    elif month in [4, 6, 9, 11]:
        if day > 30:
            return "Invalid Date"
    elif month == 2:
        if is_leap_year(year):
            if day > 29:
                return "Invalid Date"
        else:
            if day > 28:
                return "Invalid Date"

    return f"{year:04d}-{month:02d}-{day:02d}"
# Test Cases
assert validate_and_format_date("02/29/2024") == "2024-02-29"
assert validate_and_format_date("02/29/2023") == "Invalid Date"
assert validate_and_format_date("13/01/2023") == "Invalid Date"
assert validate_and_format_date("00/10/2023") == "Invalid Date"
assert validate_and_format_date("12/31/2023") == "2023-12-31"
assert validate_and_format_date("01/01/2023") == "2023-01-01"
assert validate_and_format_date("02/30/2024") == "Invalid Date"
assert validate_and_format_date("invalid") == "Invalid Date"
print("All test cases passed successfully ✅")
```

```
/usr/bin/python3 /Users/anumandlarithika/SRU/Ai_Assisted_lab/lab8.1/task05.py
anumandlarithika@ANUMANDLAs-MacBook-Air Ai_Assisted_lab % /usr/bin/python3 /Users/anumandlarithika/SRU/Ai_Assisted_lab/lab8.1/task05.py
All test cases passed successfully ✅
anumandlarithika@ANUMANDLAs-MacBook-Air Ai_Assisted_lab %
```

Explanation: This ensures a date is valid in “MM/DD/YYYY” format and neatly converts it into “YYYY-MM-DD.